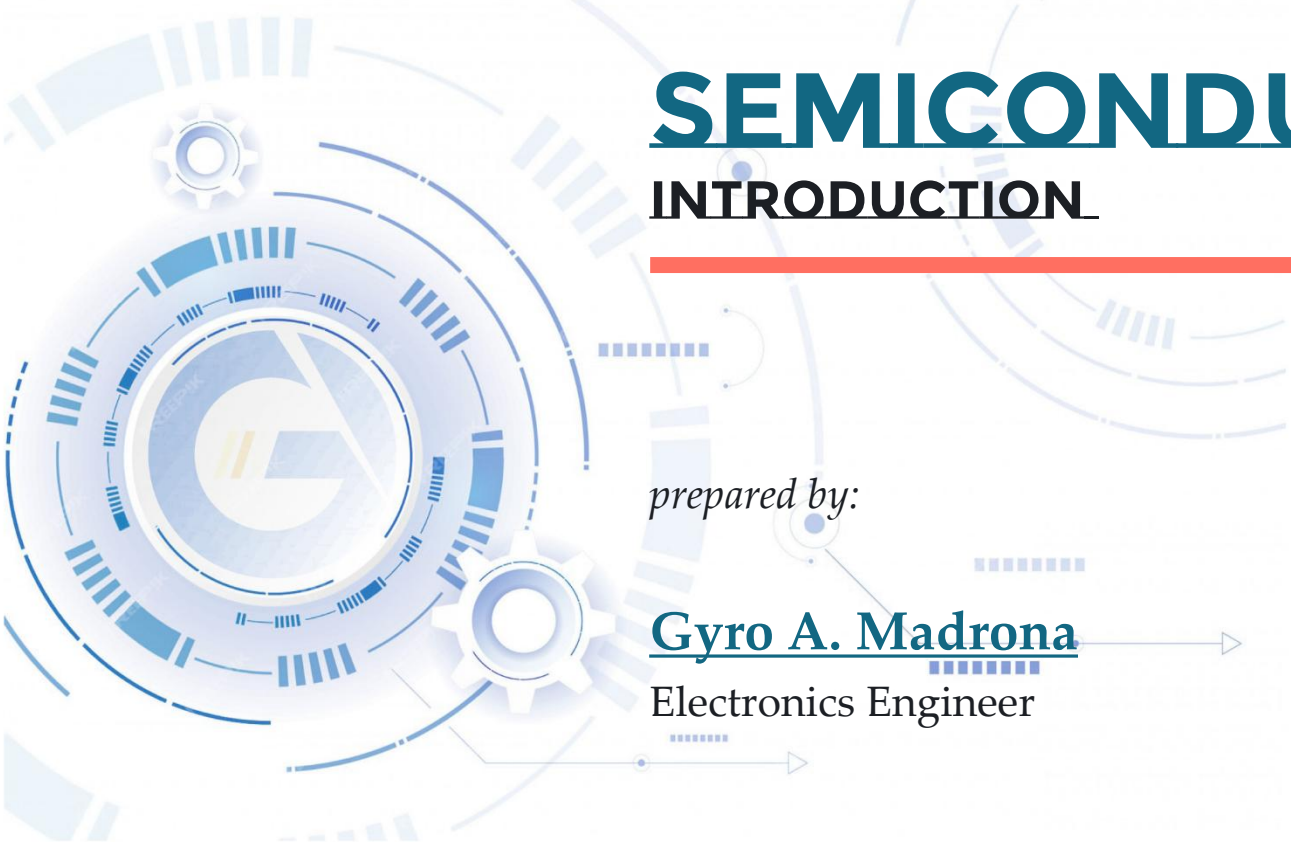




SEMICONDUCTOR

INTRODUCTION

prepared by:

Gyro A. Madrona
Electronics Engineer

TOPIC OUTLINE

Atomic Structure

Properties of Semiconductor

Two Types of Extrinsic Semiconductor



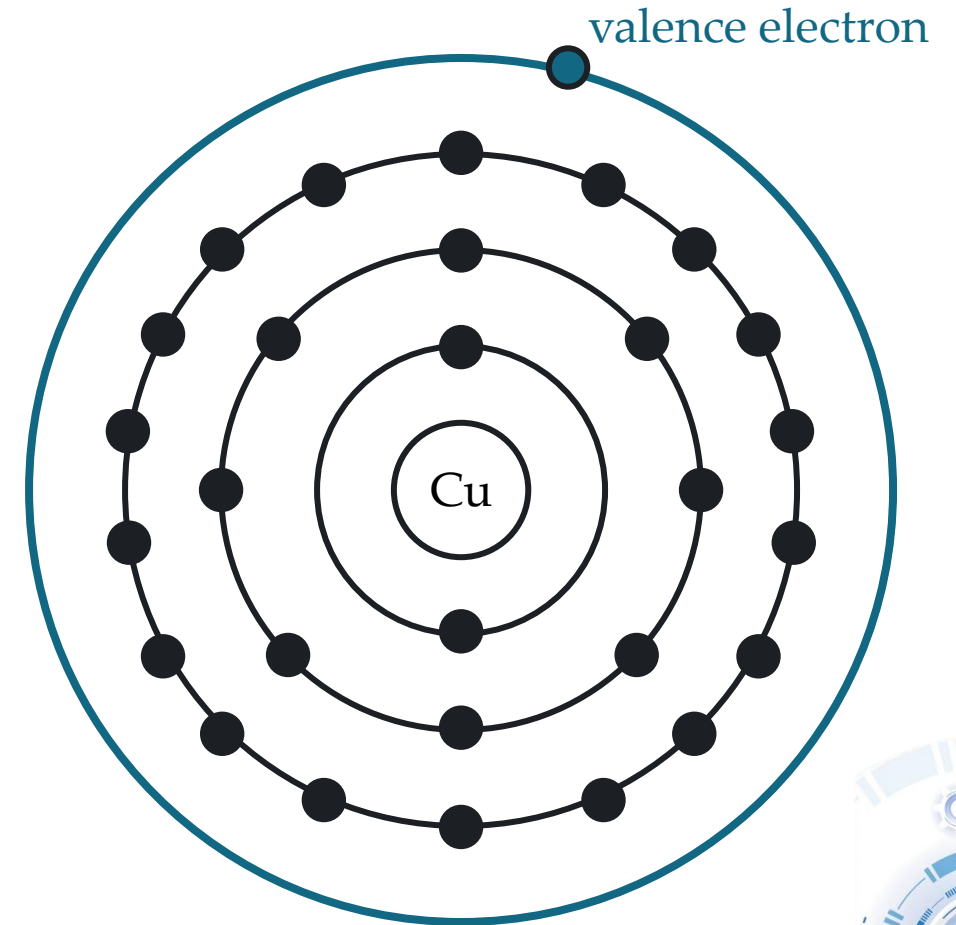
ATOMIC STRUCTURE



VALENCE ORBIT

The valence orbit (or valence shell) is the outermost electron shell of an atom. It contains the valence electron(s), which control the electrical properties of the atom.

Copper Atom



CONDUCTOR

Conductors are materials that allow the electric current to flow through them.

Valence electrons

Less than 4 electrons

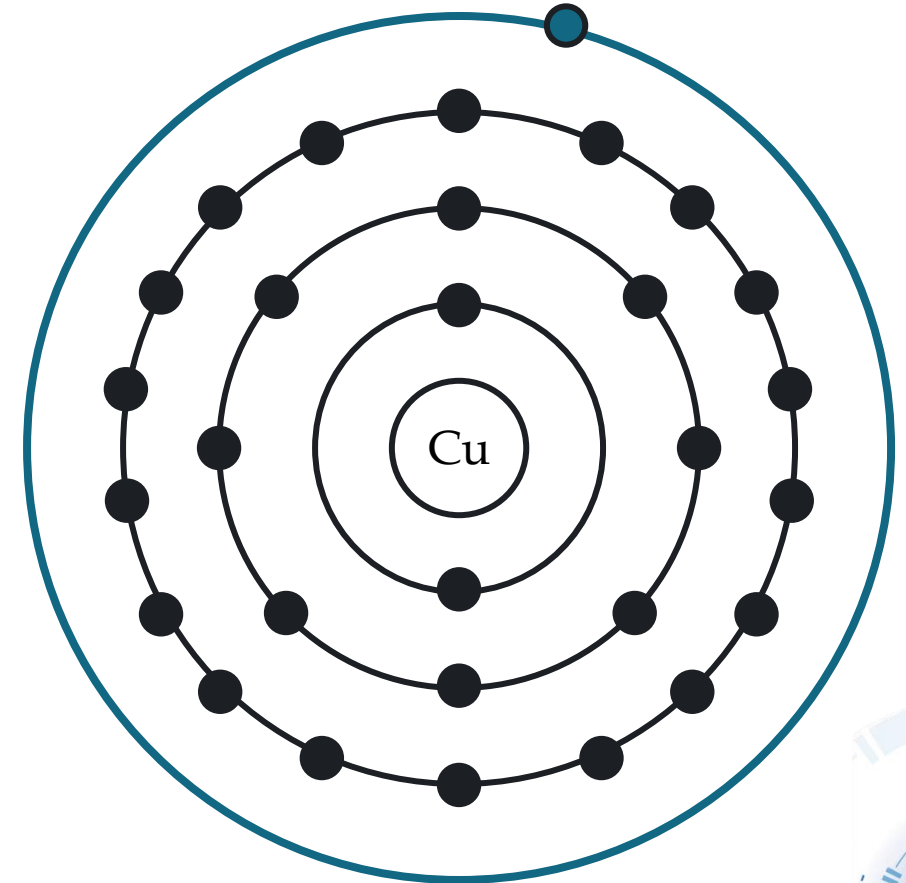
Example

Copper

Silver

Gold

Copper Atom



INSULATOR

Insulators are materials that do not conduct electricity.

Valence electrons

More than 4 electrons

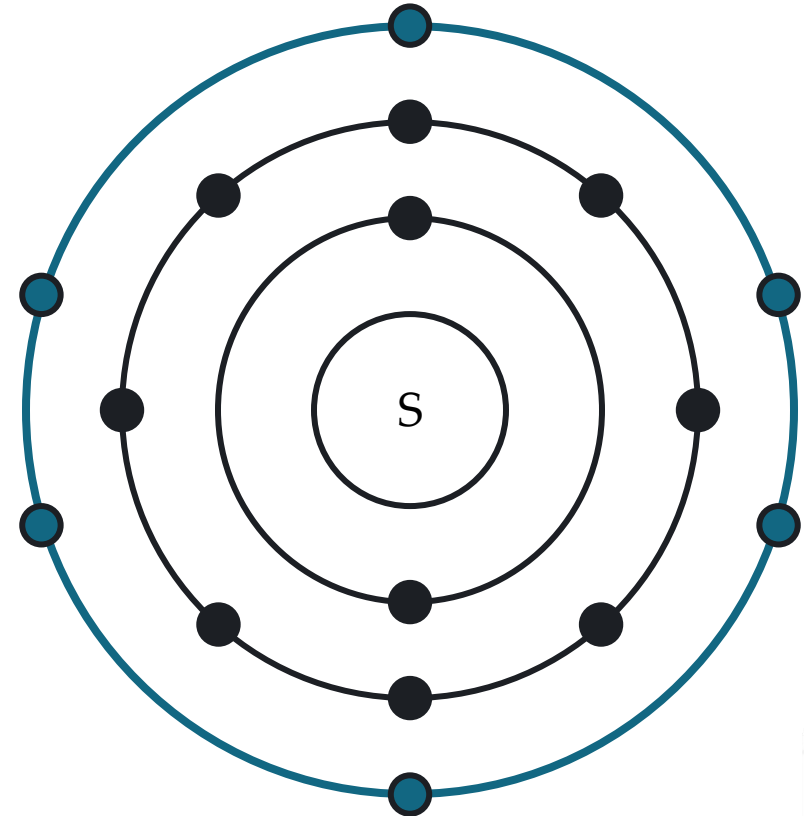
Example

Rubber

Plastic

Sulfur

Sulfur Atom



SEMICONDUCTOR

Semiconductors are materials that have an electrical conductivity between that of a conductor and an insulator.

Valence electrons

Exactly 4 electrons

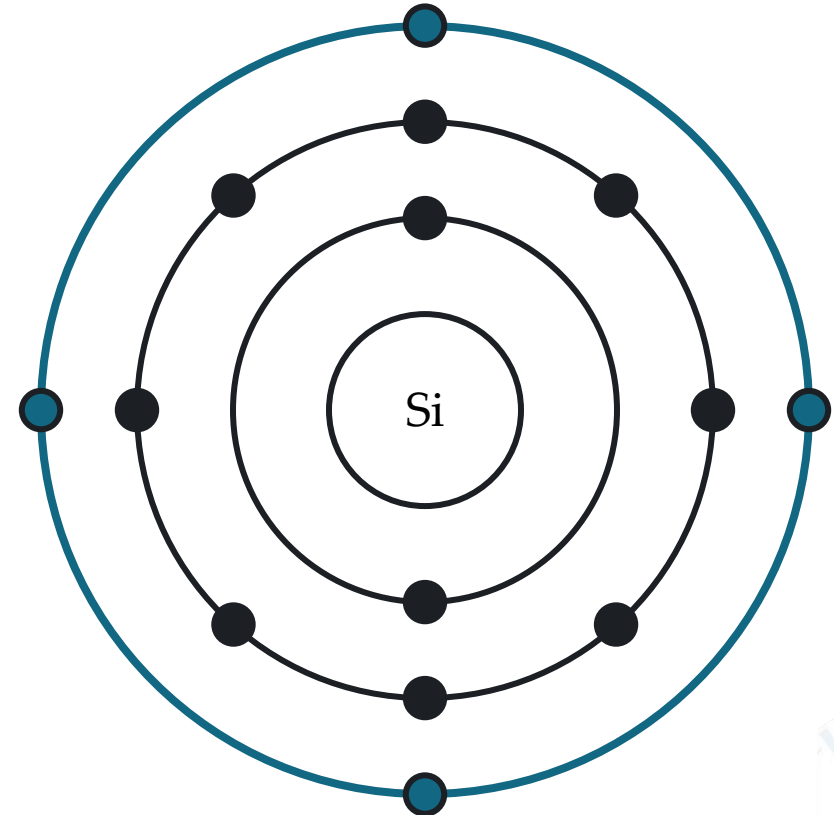
Example

Silicon

Carbon

Germanium

Silicon Atom



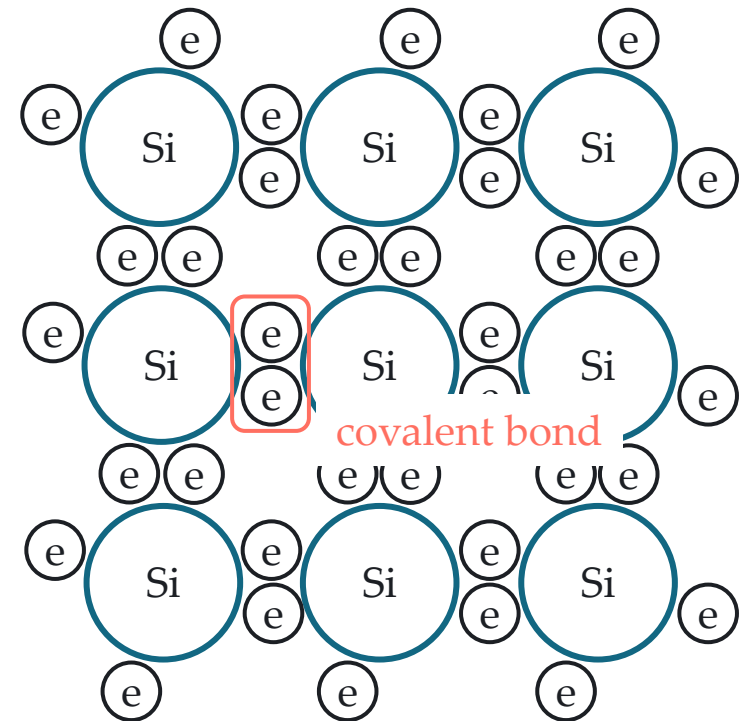
PROPERTIES OF SEMICONDUCTOR



THE SILICON ATOM

A valence saturation occurs when an atom's outermost electron shell (valence orbit) reaches its maximum capacity of 8 electrons.

Silicon Crystal

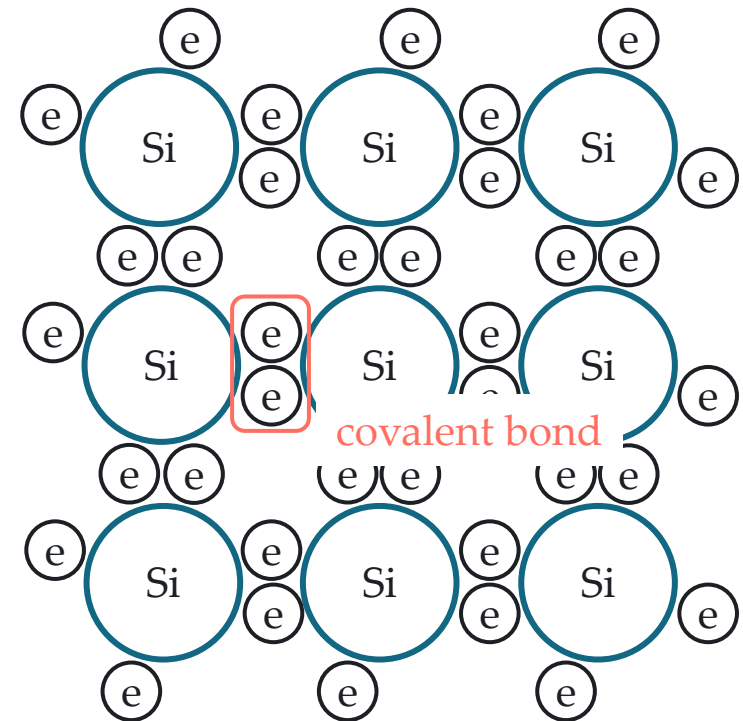


INTRINSIC SEMICONDUCTOR

An intrinsic semiconductor is a pure semiconductor.

At room temperature, it acts like an insulator.

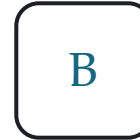
Silicon Crystal



EXTRINSIC SEMICONDUCTOR

Doping is the process of **adding** impurity atoms to an intrinsic crystal to alter its electrical conductivity.

Group III Elements



Boron



Aluminum



Gallium



Indium

Group V Elements



Phosphorus



Arsenic



Antimony



TWO TYPES OF EXTRINSIC SEMICONDUCTOR



TWO TYPES OF FLOW

Electron flow is the movement of free electrons in a semiconductor (or conductor).

Hole flow is the movement of “empty spaces” (holes) left behind when electrons jump in the valence band.

Electron Flow



Hole Flow



N-TYPE SEMICONDUCTOR

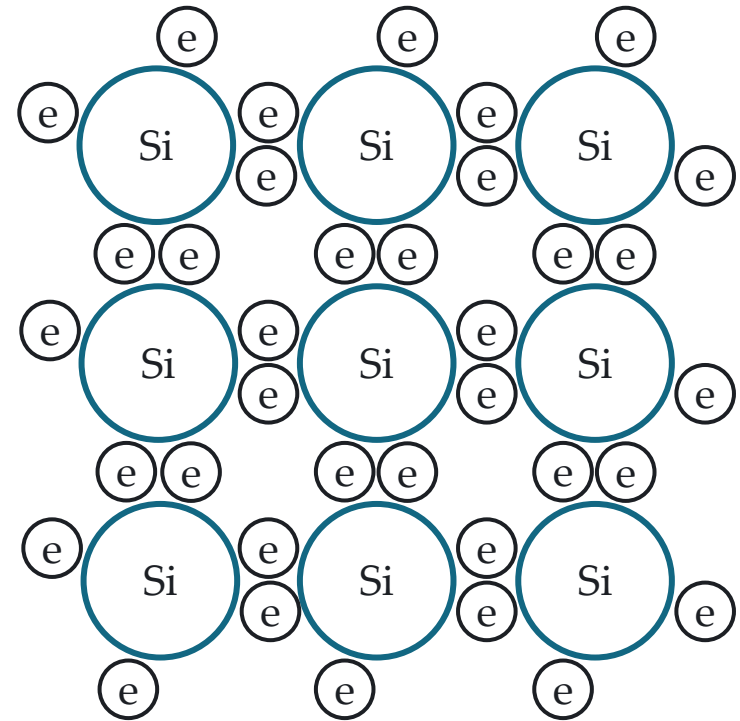
An **n-type** semiconductor is created by doping intrinsic semiconductor with a **pentavalent** impurity.
example

Arsenic (As)

Antimony (Sb)

Phosphorus (P)

Silicon Crystal



N-TYPE SEMICONDUCTOR

An **n-type** semiconductor is created by doping intrinsic semiconductor with a **pentavalent** impurity.

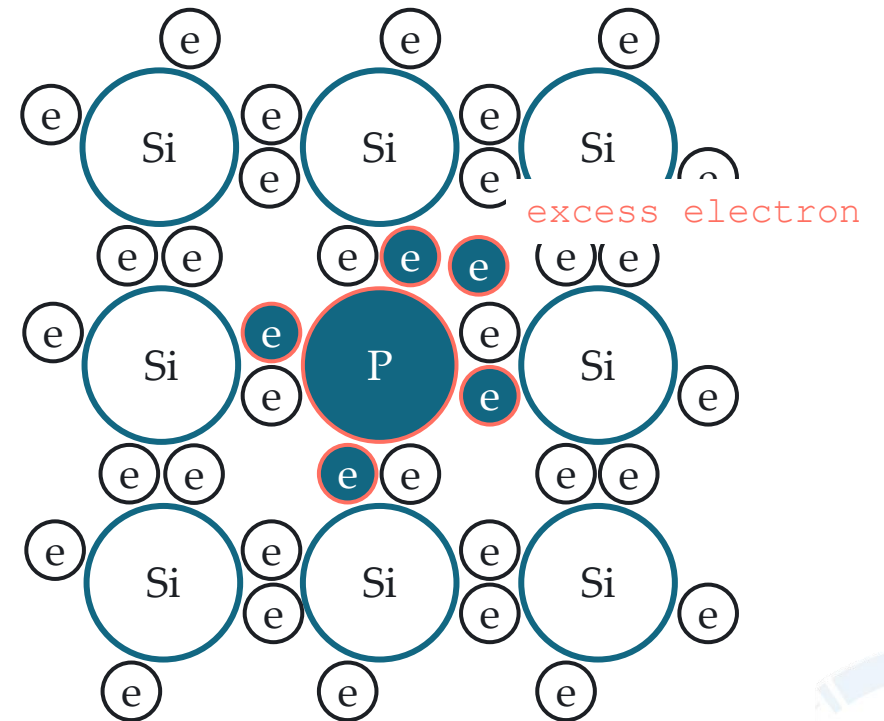
Example

Arsenic (As)

Antimony (Sb)

Phosphorus (P)

Doped with Phosphorus



N-TYPE SEMICONDUCTOR

An **n-type** semiconductor is created by doping intrinsic semiconductor with a **pentavalent** impurity.

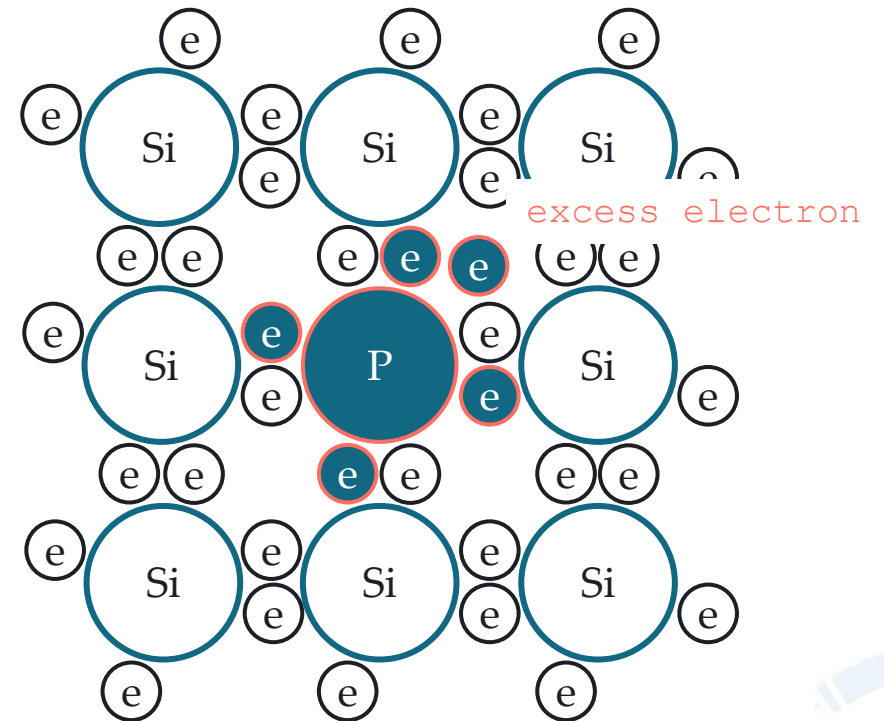
Majority carriers

Electrons

Minority carriers

Holes

Doped with Phosphorus



P-TYPE SEMICONDUCTOR

A **p-type** semiconductor is created by doping intrinsic semiconductor with a **trivalent** impurity.

Example

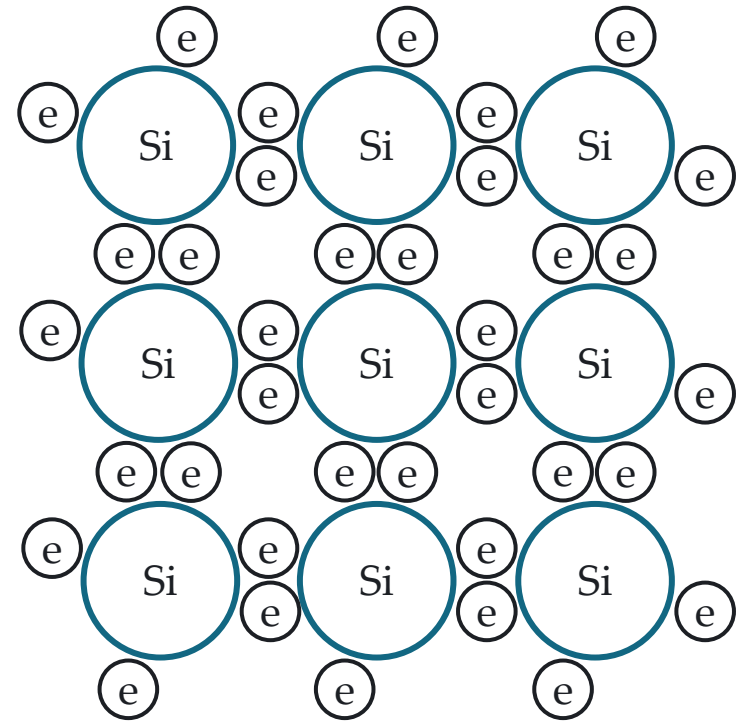
Aluminum (Al)

Boron (B)

Gallium (Ga)

Indium (In)

Silicon Crystal



P-TYPE SEMICONDUCTOR

A **p-type** semiconductor is created by doping intrinsic semiconductor with a **trivalent** impurity.

Example

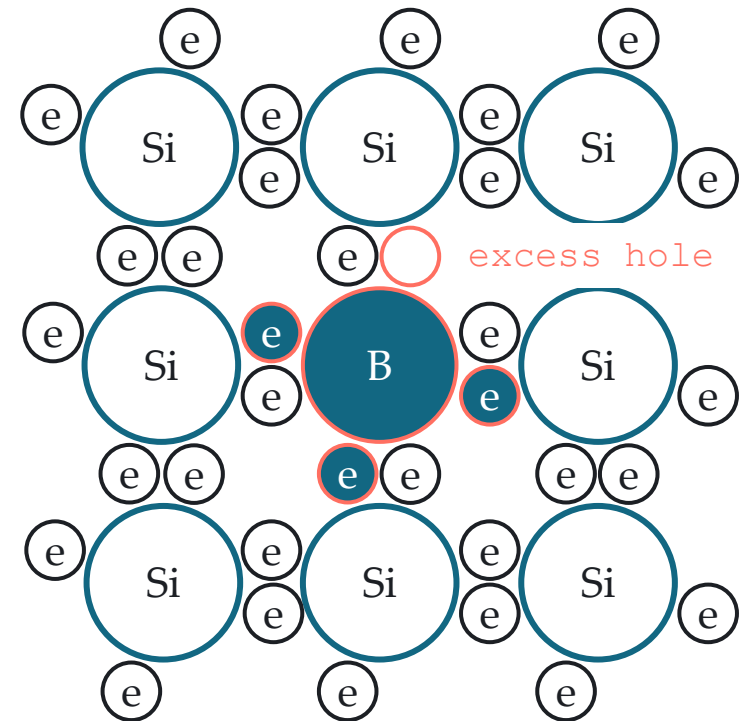
Aluminum (Al)

Boron (B)

Gallium (Ga)

Indium (In)

Doped with Boron



P-TYPE SEMICONDUCTOR

A **p-type** semiconductor is created by doping intrinsic semiconductor with a **trivalent** impurity.

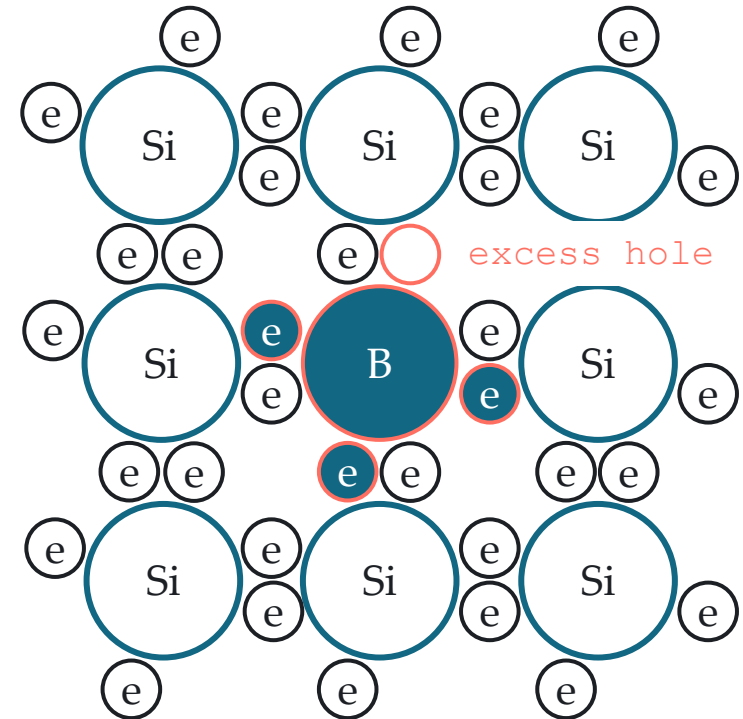
Majority carriers

Holes

Minority carriers

Electrons

Doped with Boron



EXERCISE

A doped semiconductor has 10 billion silicon atoms and 15 million pentavalent atoms. If the ambient temperature is 25°C , how many free electrons and holes are there inside the semiconductor?

Solution



EXERCISE

In reference to the previous example, if 5 million trivalent atoms are added instead of pentavalent atoms, how many holes are there inside the semiconductor?

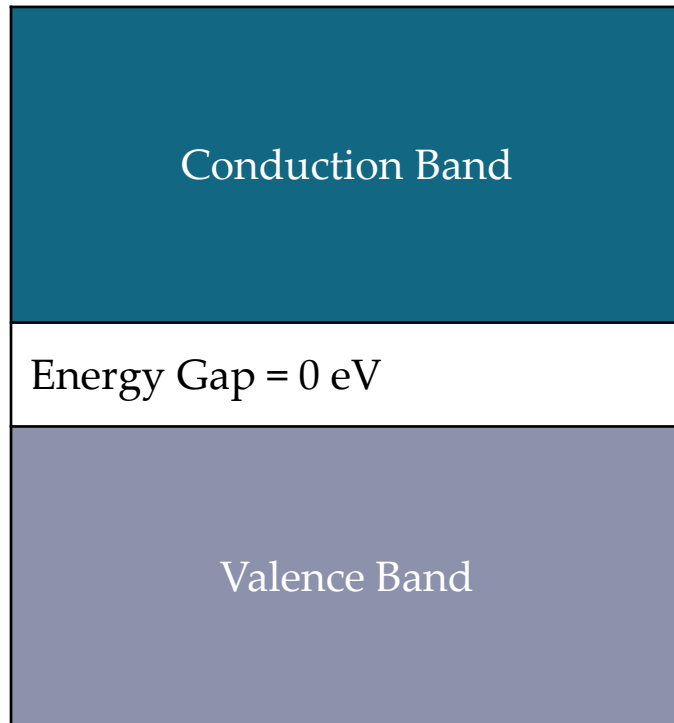
Solution



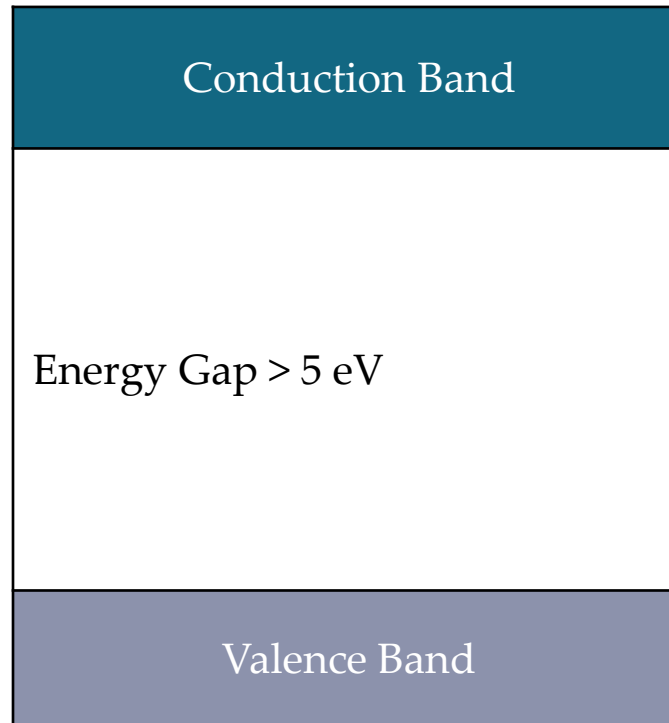
ENERGY LEVELS

Electronvolt (eV) is the energy needed to move one electron through a potential difference of 1 volt.

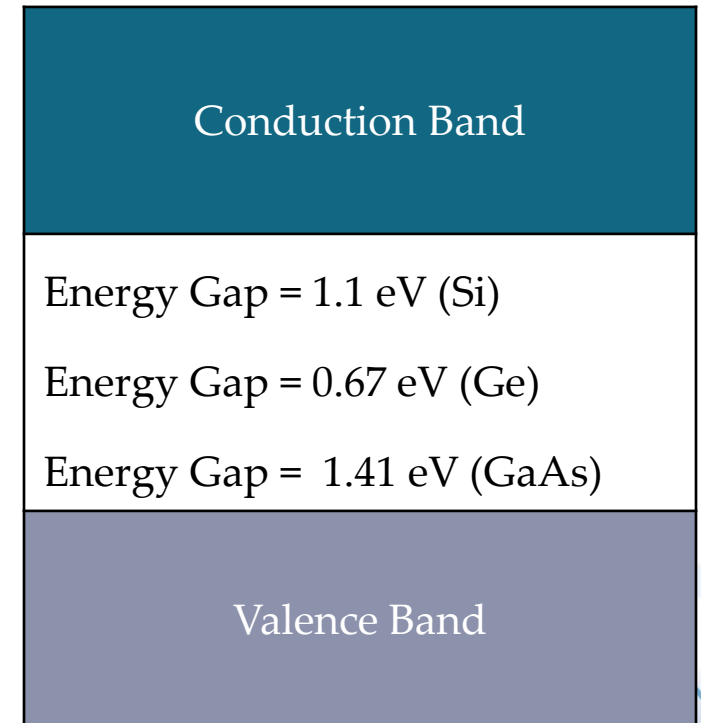
Conductor



Insulator



Semiconductor



LABORATORY

